

HW 7

Townsend 7.1 and 7.3 (5 pts)

Problem 7.1 Show that the constant $c_- = \sqrt{n}$ in (7.24), that is,

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$$

using the procedure we used to establish that $c_+ = \sqrt{n+1}$, that is,

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

We can write

$$\begin{aligned}\langle n|\hat{a}^\dagger\hat{a}|n\rangle &= c_-^*c_- \langle n-1|n-1\rangle \\ n &= c_-^*c_- \\ c_- &= \sqrt{n} \\ \hat{a}|n\rangle &= \sqrt{n}|n-1\rangle\end{aligned}$$

Problem 7.3 Show that properly normalized eigenstates of the harmonic oscillator are given by (7.37). *Suggestion:* Use induction.

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle$$

For $n = 0$,

$$|0\rangle = 1|0\rangle$$

For $n = 1$

$$\begin{aligned}|n\rangle &= \frac{a^\dagger}{\sqrt{1!}}|0\rangle \\ &= \frac{1}{\sqrt{1!}}|0+1\rangle \\ &= 1|1\rangle\end{aligned}$$

For an arbitrary n greater than 1

$$\begin{aligned}\frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle &= \sqrt{1}\frac{(a^\dagger)^{n-1}}{\sqrt{n!}}|1\rangle \\ &= \sqrt{2}\sqrt{1}\frac{(a^\dagger)^{n-2}}{\sqrt{n!}}|2\rangle \\ &= \frac{\sqrt{k!}}{\sqrt{n!}}(a^\dagger)^{n-k}|k\rangle\end{aligned}$$

for the special case of $k = n$, this gets up to

$$\begin{aligned}\frac{\sqrt{n!}}{\sqrt{n}}(a^\dagger)^0|n\rangle \\ = |n\rangle\end{aligned}$$

Townsend Problem 7.7 (5 pts)

A particle of mass m in the one-dimensional harmonic oscillator is in a state for which a measurement of the energy yields the values $\hbar\omega/2$ or $3\hbar\omega/2$, each with a probability of one-half. The average value of the momentum $\langle p_x \rangle$ at time $t = 0$ is $\sqrt{m\omega\hbar/2}$. This information specifies the state of the particle completely. What is this state and what is $\langle p_x \rangle$ at time t ?

state has $|0\rangle$ and $|1\rangle$ superposition, and we'll have a cos something for expectation value
we have the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle)$$

We can use the average value of momentum at time zero to work out the phase difference between $|0\rangle$ and $|1\rangle$.

We can write $\hat{p}_x$ in terms of $\hat{a}$ and $\hat{a}^\dagger$.

$$\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}}(\hat{a} - \hat{a}^\dagger)$$

If $\langle \hat{p}_x \rangle = 0$ at $t = 0$, we can write

$$\begin{aligned} 0 &= \langle \psi | \hat{p}_x | \psi \rangle \\ &= \frac{1}{\sqrt{2}}(\langle 0 | + \langle 1 |) \hat{p}_x (|0\rangle + |1\rangle) \\ &= \frac{1}{\sqrt{2}}((\langle 0 | + \langle 1 |) \hat{p}_x |0\rangle + (\langle 0 | + \langle 1 |) \hat{p}_x |1\rangle) \\ &= \frac{-1}{\sqrt{2}}\sqrt{\frac{m\omega\hbar}{2}}\left((\langle 0 | + \langle 1 |)(\hat{a} - \hat{a}^\dagger)|0\rangle + (\langle 0 | + \langle 1 |)(\hat{a} - \hat{a}^\dagger)|1\rangle\right) \end{aligned}$$

$$\begin{aligned} \hat{a}|n\rangle &= \sqrt{n}|n-1\rangle \\ \hat{a}^\dagger|n\rangle &= \sqrt{n+1}|n+1\rangle \end{aligned}$$

$$\begin{aligned} \hat{a}|0\rangle &= 0 \\ \hat{a}|1\rangle &= |0\rangle \\ \hat{a}^\dagger|0\rangle &= |1\rangle \\ \hat{a}^\dagger|1\rangle &= \sqrt{2}|2\rangle \end{aligned}$$

we can decompose the terrible

$$\begin{aligned} &\left((\langle 0 | + \langle 1 | e^{-i\phi})(\hat{a} - \hat{a}^\dagger)|0\rangle + (\langle 0 | + \langle 1 | e^{-i\phi})(\hat{a} - \hat{a}^\dagger)|1\rangle e^{i\phi}\right) \\ &(\langle 0 | + \langle 1 | e^{-i\phi})(-1|1\rangle) + (\langle 0 | + \langle 1 | e^{-i\phi})(|0\rangle e^{i\phi} - \sqrt{2}|2\rangle e^{i\phi}) \end{aligned}$$

Because we have the common $\frac{1}{\sqrt{2}}$ terms on each bra and ket pair, we'll get

we have an $e^{i\phi}$ from the $|1\rangle$ term that went down to $|0\rangle$, and an $e^{-i\phi}$ term from the $|0\rangle$ that went up to $|1\rangle$ and now hit the $\langle 1 | e^{-i\phi}$ term - the others are all orthogonal

$$\frac{1}{2}(e^{i\phi} - e^{-i\phi})$$

Which looks like

$$i \sin \phi$$

Therefore, we have

$$\langle \hat{p}_x \rangle = \sqrt{\frac{m\omega\hbar}{2}} \sin \phi$$

at time $t = 0$ is $\sqrt{m\omega\hbar/2}$.

So we have

$\sin \phi = 1$ since there is no time dependence in the expectation value at $t = 0$ (??? this feels weird)

$\phi = \frac{\pi}{2}$.

The state is therefore

$$\begin{aligned} & \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}e^{i\frac{\pi}{2}}|1\rangle \\ &= \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle \end{aligned}$$

We can time evolve the state with

$$\begin{aligned} & e^{-i\frac{E}{\hbar}t} \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle \right) \\ &= \frac{1}{\sqrt{2}}e^{i\frac{\omega}{2}t}|0\rangle + \frac{i}{\sqrt{2}}e^{i\frac{3}{2}\omega t}|1\rangle \\ &= \frac{1}{\sqrt{2}}e^{i\frac{\omega}{2}t} (|0\rangle + ie^{i\omega t}|1\rangle) \end{aligned}$$

We can just tack on this phase to our expectation value calculation from earlier, to get

$$\langle \hat{p}_x \rangle(t) = \sqrt{\frac{m\omega\hbar}{2}} \sin \left(\frac{\pi}{2} + \omega t \right)$$

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p}_x \right) \quad (7.8)$$

and

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p}_x \right) \quad (7.9)$$

in a fashion similar to the way we introduced $\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y$. Since $\hat{x}$ and $\hat{p}_x$ have different dimensions, we cannot just add $\hat{x} \pm i\hat{p}_x$, as we did for angular momentum. The factor of $\sqrt{m\omega/2\hbar}$ is inserted in front so that the operators (7.8) and (7.9) are dimensionless. Using (7.7), we can verify that these operators satisfy the simple commutation relation

$$[\hat{a}, \hat{a}^\dagger] = 1 \quad (7.10)$$

Inverting (7.8) and (7.9), we obtain

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \quad (7.11)$$

and

$$\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a} - \hat{a}^\dagger) \quad (7.12)$$

which can be used to express the Hamiltonian (7.6) as

$$\hat{H} = \frac{\hbar\omega}{2} (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger) = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (7.13)$$

Townsend Problem 7.9 (5 pts)

Show that in the superposition of adjacent energy states (7.63) the average value of the position of the particle is given by

$$\langle x \rangle = \langle \psi | \hat{x} | \psi \rangle = A \cos(\omega t + \delta)$$

and the average value of the momentum is given by

$$\langle p_x \rangle = \langle \psi | \hat{p}_x | \psi \rangle = -m\omega A \sin(\omega t + \delta)$$

in accord with Ehrenfest's theorem, (6.33) and (6.34).

Using (4.16), we can calculate the time rate of change of the expectation value of the position of the particle:

$$\begin{aligned} \frac{d\langle x \rangle}{dt} &= \frac{i}{\hbar} \langle \psi | [\hat{H}, \hat{x}] | \psi \rangle = \frac{i}{2m\hbar} \langle \psi | [\hat{p}_x^2, \hat{x}] | \psi \rangle \\ &= \frac{i}{2m\hbar} \langle \psi | (\hat{p}_x [\hat{p}_x, \hat{x}] + [\hat{p}_x, \hat{x}] \hat{p}_x) | \psi \rangle \\ &= \frac{\langle \psi | \hat{p}_x | \psi \rangle}{m} \frac{\langle \hat{p}_x \rangle}{m} \end{aligned} \quad (6.33)$$

Moreover, you may also check (see Problem 6.1) that

$$\begin{aligned} \frac{d\langle p_x \rangle}{dt} &= \frac{i}{\hbar} \langle \psi | [\hat{H}, \hat{p}_x] | \psi \rangle \\ &= \left\langle \frac{dV}{dx} \right\rangle \end{aligned} \quad (6.34)$$

We can apply Ehrenfest's theorem to see if the relation holds.

$$\frac{d}{dt} \langle x \rangle \stackrel{?}{=} \frac{1}{m} \langle p_x \rangle$$

$$-A\omega \sin(\omega t + \delta) = -m\omega \frac{A}{m} \sin(\omega t + \delta)$$

and

$$\frac{d}{dt} \langle p_x \rangle \stackrel{?}{=} \frac{i}{\hbar} \langle \psi | [\hat{H}, \hat{p}_x] | \psi \rangle$$

$$\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a} - \hat{a}^\dagger) \quad (7.12)$$

which can be used to express the Hamiltonian (7.6) as

$$\hat{H} = \frac{\hbar\omega}{2} (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger) = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (7.13)$$

The commutation between $\hat{H}$ and $\hat{p}_x$ is

$$\begin{aligned} &\left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) (\hat{a} - \hat{a}^\dagger) - (\hat{a} - \hat{a}^\dagger) \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \\ &= \hat{a}^\dagger \hat{a} (\hat{a} - \hat{a}^\dagger) - (\hat{a} - \hat{a}^\dagger) (\hat{a}^\dagger \hat{a}) \\ &= \hat{a}^\dagger \hat{a} \hat{a} - \hat{a}^\dagger \hat{a} \hat{a}^\dagger - (\hat{a} \hat{a}^\dagger \hat{a} - \hat{a}^\dagger \hat{a}^\dagger \hat{a}) \\ &= \hat{a}^\dagger \hat{a} \hat{a} - \hat{a}^\dagger \hat{a} \hat{a}^\dagger - \hat{a} \hat{a}^\dagger \hat{a} + \hat{a}^\dagger \hat{a}^\dagger \hat{a} \\ &= \hat{a}^\dagger (\hat{a} \hat{a} + \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger) - \hat{a} \hat{a}^\dagger \hat{a} \end{aligned}$$

We know that $[\hat{a}, \hat{a}^\dagger] = 1$, and conversely that $[\hat{a}^\dagger, \hat{a}] = -1$

$$\begin{aligned} &\hat{a}^\dagger (\hat{a} \hat{a} - 1) - \hat{a} \hat{a}^\dagger \hat{a} \\ &= \hat{a}^\dagger \hat{a} \hat{a} - \hat{a} \hat{a}^\dagger \hat{a} - \hat{a}^\dagger \end{aligned}$$

which looks like

$$\begin{aligned} &(\hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger) \hat{a} - \hat{a}^\dagger \\ &= -\hat{a} - \hat{a}^\dagger \\ &= -(\hat{a} + \hat{a}^\dagger) \end{aligned}$$

With constants, we have

$$[\hat{H}, \hat{p}_x] = -(\hbar\omega) \left(i\sqrt{\frac{m\omega\hbar}{2}} \right) (\hat{a} + \hat{a}^\dagger)$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \quad (7.11)$$

and

$$\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a} - \hat{a}^\dagger) \quad (7.12)$$

$$\hat{x}\sqrt{\frac{2m\omega}{\hbar}} = (\hat{a} + \hat{a}^\dagger)$$

so

$$\begin{aligned} [\hat{H}, \hat{p}_x] &= -i\hbar\omega\sqrt{\frac{m\omega\hbar}{2}}\sqrt{\frac{2m\omega}{\hbar}}\hat{x} \\ &= -i\hbar\omega m\omega\hat{x} \\ &= -i\hbar m\omega^2\hat{x} \end{aligned}$$

Finally we can check

$$\langle x \rangle = \langle \psi | \hat{x} | \psi \rangle = A \cos(\omega t + \delta)$$

the momentum is given by

$$\langle p_x \rangle = \langle \psi | \hat{p}_x | \psi \rangle = -m\omega A \sin(\omega t + \delta)$$

$$\frac{d}{dt} \langle p_x \rangle \stackrel{?}{=} \frac{i}{\hbar} \langle \psi | [\hat{H}, \hat{p}_x] | \psi \rangle$$

because we have $\frac{i}{\hbar}$, the $i\hbar$ just goes to -1 .

$$\frac{d}{dt} \langle p_x \rangle \stackrel{?}{=} -m\omega^2 \langle \hat{x} \rangle$$

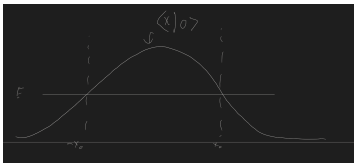
$$\frac{d}{dt} \langle p_x \rangle = m\omega^2 A \cos(\omega t + \delta)$$

Which is indeed $m\omega^2 \langle \hat{x} \rangle$

Thus, both parts of Ehrenfest's theorem are satisfied.

Townsend Problem 7.13 (5 pts)

Calculate the probability that a particle in the ground state of the harmonic oscillator is located in a classically disallowed region, namely, where $V(x) > E$. Obtain a numerical value for the probability. *Suggestion:* Express your integral in terms of a dimensionless variable and compare with the tabulated values of the error function.



We have the ground state in a simple harmonic oscillator given as

$$\langle x|0 \rangle = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-m\omega \frac{x^2}{2\hbar}}$$

from pg 255 eq 7.44

We can find the probability of landing within this area as

$$\int_{-x_0}^{x_0} \psi^* \psi$$

$$= \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{2}} \int_{-x_0}^{x_0} e^{-m\omega \frac{x^2}{\hbar}} dx$$

For the SHO, we use a potential function

$$\frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2$$

We know that $E = \hbar\omega \left(n + \frac{1}{2} \right)$, so
for the ground state $E = \frac{\hbar\omega}{2}$.

We can solve for where

$$\frac{1}{2} m\omega^2 x^2 = \hbar \frac{\omega}{2}$$

$$x = \sqrt{\frac{\hbar}{m\omega}}$$

I couldn't figure out the Usub and didn't know what the Error Function was, so I took a hit and looked up the integral on symbolab, knowing that I would dock points for it. Sorry! I am running out of time to submit and didn't want to loose everything else for not figuring out an integral that was easily computable online.

$$= \frac{1}{2} \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{2}} \left(\sqrt{\frac{\pi\hbar}{m\omega}} \right) \operatorname{erf} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) \Big|_{-x_0}^{x_0}$$

$$= \frac{1}{2} \operatorname{erf} \left(\sqrt{\frac{m\omega}{\hbar}} x \right) \Big|_{-x_0}^{x_0}$$

$$= \frac{1}{2} \left(\operatorname{erf} \left(\underbrace{\sqrt{\frac{m\omega}{\hbar}} \sqrt{\frac{\hbar}{m\omega}}}_1 \right) - \operatorname{erf}(-1) \right)$$

$$\operatorname{erf}(1) = 0.842$$

$$\operatorname{erf}(-1) = -0.842$$

So this just gets to
 $P(\text{ allowed }) = 0.842$
thus,
 $P(\text{ not classical }) = 0.158$

Townsend Problem 7.14 (5 pts)

As shown in Section 7.1, for small oscillations a pendulum behaves as a simple harmonic oscillator. Suppose a particle of mass m is in the ground state of a pendulum of length L and that instantaneously the length of the pendulum increases from L to $4L$. What is the probability the particle will be found to be in the ground state of this new oscillator. Give a numerical answer.

The ground state in each well is given by

$$\langle x|0 \rangle = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-m\omega \frac{x^2}{2\hbar}}$$

If we call the initial state $|i\rangle$ and the final state $|f\rangle$, then this is the same as asking

$$\langle f|i \rangle$$

$$\int_{-\frac{L}{2}}^{\frac{L}{2}} dx \langle f|x \rangle \langle x|i \rangle$$

because the initial function is zero outside the length L .

For the first ground state,

$$\omega_0 = \sqrt{\frac{g}{L}}$$

For the second ground state,

$$\begin{aligned}\omega &= \sqrt{\frac{g}{4L}} \\ &= \frac{1}{2} \sqrt{\frac{g}{L}} \\ &= \frac{1}{2} \omega_0\end{aligned}$$

Therefore, we have

$$\begin{aligned}&\int_{-\frac{L}{2}}^{\frac{L}{2}} dx \left(\frac{m\omega_0}{\pi\hbar} \right)^{\frac{1}{4}} \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-m\omega_0 \frac{x^2}{2\hbar}} e^{-m\omega \frac{x^2}{2\hbar}} \\ &= \left(\frac{m^2\omega\omega_0}{\pi^2\hbar^2} \right)^{\frac{1}{4}} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx e^{-m \frac{x^2}{2\hbar} (\omega_0 + \omega)}\end{aligned}$$

Using the appendix,

$$\int_{-\infty}^{\infty} e^{-ax^2} = \sqrt{\frac{\pi}{a}}$$

Here, $a = \frac{m}{2\hbar}(\omega_0 + \omega)$

So we have

$$\begin{aligned}&\left(\frac{m^2\omega\omega_0}{\pi^2\hbar^2} \right)^{\frac{1}{4}} \sqrt{\frac{2\hbar\pi}{m(\omega + \omega_0)}} \\ &= \left(\frac{1}{2\pi^2\hbar^2} m^2 \frac{1}{2} \omega_0^2 \right)^{\frac{1}{4}} \left(\frac{4\hbar^2\pi^2}{m^2 \left(\frac{3}{2}\omega_0 \right)^2} \right)^{\frac{1}{4}} \\ &= \frac{2}{3}\end{aligned}$$